

The EVOLUTION Of LCD DISPLAYS And HOW It MATTERS

In a competitive market where fast designing is needed, how far will liquid crystal displays (LCDs) be an option for design engineers? Let's find out

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Ten years ago, my senior told me that the LCD is going to die, but it has paved its way into our everyday life, starting from our phones to our vehicles. Segment LCDs were simple in design where an input channel was given directly to display digits. Even if videos could not be displayed, these LCDs were used on old auto meters and were pre-designed. Though a little complicated in design, thin film transistor liquid crystal displays (TFT LCDs) are becoming quite popular in smartphones and vehicles nowadays.

Liquid crystal can guide the light. For a TN LCD when the transistor of the TFT circuit is turned off and the LC changes the orientation of light rotation. After light passes through both the polarizers, we see the LCD completely white. However, when the circuit is turned on, the polarizer blocks the light, resulting in a black image. Hence liquid crystal is the core of the design that can direct the light as per its rotation.

Modes

Based on the liquid crystal rotation, there are three modes, namely, In Plane Switching (IPS), Vertical Alignment (VA), and Twisted Nematic (TN) mode. In the IPS mode, the crystal switches horizontally and in the VA mode it switches vertically. Both have a wider viewing angle and a larger colour coverage but are high in terms of cost. The best performance can be obtained in the IPS mode. The TN mode has a narrow viewing angle with a small colour coverage, but it has the largest market as it is very cheap.

Fig. 3 shows the comparison for the same photo on two different displays when the liquid crystal is rotated in the IPS and TN modes. The performance varies as the angle changes. We notice that IPS has a better performance.

Thin film transistor LCD

It is a display panel that uses liquid crystal to control the amount of light that comes from the backlight. The TFT acts like a switch that controls the liquid crystal. Each segment of liquid crystal is like a shutter that either blocks or allows light to pass through. Fig. 4 shows a TFT LCD where the top is an open cell and the bot-

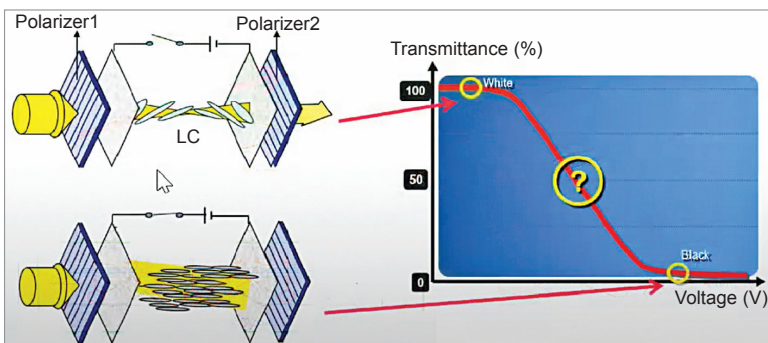


Fig. 1: Working of a liquid crystal

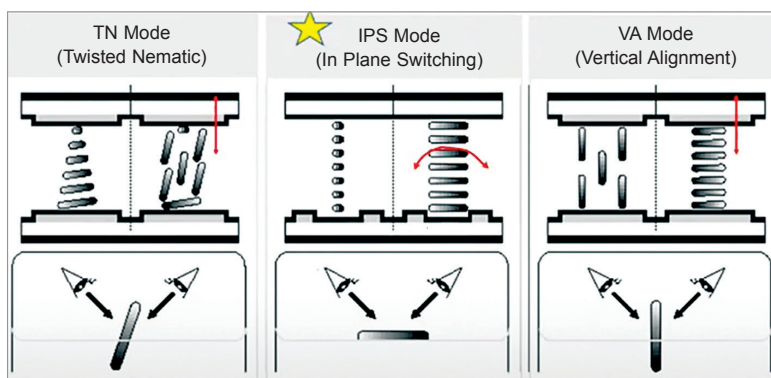


Fig. 2: Three modes of liquid crystal rotation